Hierarchical Projection as a Validity Layer for ESG Promise Verification: Multilingual Evidence

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ABSTRACT

ESG promise verification requires jointly predicted fields to satisfy hard parent–child constraints, particularly when outputs are intended for audit-oriented use. We study hierarchical projection as a retraining-free validity layer for field-wise classifiers. Seven decision rules are compared on identical cross-fitted probabilities and test rows in Chinese, followed by projection-versus-argmax replication across English, French, Japanese, and Korean corpora. Projection reduces the Chinese invalid-tuple rate from 12.55% to zero by construction. Its official weighted macro-F1 contrast is undetected ($\Delta = -0.001$, $p_{\text{Holm}} = 1.000$), while pre-specified tuple accuracy improves by +0.035 ($p_{\text{Holm}} = .001$). Across 32 external backbone-by-loss arms, tuple accuracy improves in all 32, whereas weighted macro-F1 improves in 14. These results position hierarchical projection not as a guaranteed route to higher field-wise scores, but as a practical way to guarantee coherent outputs and repeatedly improve complete-output correctness, with field-score trade-offs varying across models and corpora. Validity alone does not settle which structured rule to use: searching all 17 legal states instead of projecting greedily costs −0.006 tuple accuracy ($p_{\text{Holm}} = .028$), an adverse pre-specified result we report in full.

KEYWORDS

ESG, promise verification, hierarchical classification, constrained decoding, evaluation metrics

1 INTRODUCTION

We participated in the VeriPromiseESG task of AI CUP 2026 [1, 8], Taiwan's national inter-collegiate artificial intelligence competition, held as the AI CUP special session of NTCIR-19. This paper reports our own analysis of that task; the task definition, corpus construction and official evaluation are the organizers' and are not restated here. All results below are computed on the labeled development release, because the competition test labels are not available to participants.

Environmental, social and governance (ESG) promise verification is intended to support audit-oriented decisions, for which internal consistency is a basic requirement. Promise status, verification timeline, evidence status, and evidence quality describe one claim, and their parent–child implications determine whether the combined output is usable as a coherent record. A paragraph predicted to contain no promise cannot coherently receive a substantive timeline, and a prediction of no evidence cannot coherently receive an evidence quality. Individually plausible field labels can therefore form an invalid tuple.

Field-wise weighted macro-F1 is a standard measure of classification quality for multi-field tasks, and it is the official ranking metric here. It does not, however, assess whether the four predictions form a jointly legal tuple. A validity-preserving rule may therefore repair an unsupported child without receiving a corresponding field-score gain. The metric places no weight at all on legality, and the consequence is larger than that phrasing suggests: because each field is scored on its own, a correct field still earns its weight even when the ancestors that would license it are wrong, so an invalid tuple retains 44.3% of the available weighted field credit on average. Table 5 derives that figure term by term. We ask whether a retraining-free inference constraint can guarantee legal outputs, and how its effects differ under field-wise and whole-tuple evaluation.

We answer that question through a controlled decision comparison rather than a model or competition-performance claim. Seven rules consume identical cross-fitted base probabilities and test rows. Projection acts as a validity layer over those probabilities: it guarantees a legal tuple without changing the encoder or retraining the classifier. In Chinese, the confirmatory official-score family is null, yet pre-specified exact whole-tuple accuracy improves under projection. Across four ML-Promise languages, four backbones per language, and two fixed training objectives, tuple accuracy improves in all 32 external arms while weighted macro-F1 has a mixed sign. These are within-corpus decision contrasts; raw scores do not isolate language effects.

Our contributions are:

(1) We formalize the supplied hierarchy as 17 legal tuples out of 120 and verify that all 2,000 audited Chinese paragraphs from 49 companies are legal.
(2) We evaluate projection as a retraining-free validity layer in a seven-rule controlled comparison, reporting the complete confirmatory official-score family and the pre-specified tuple family, including adverse results.
(3) We externally replicate the within-arm projection contrast across four languages and all 32 external arms, reported one

per row in Table 7, while keeping post-hoc metrics and structural and backbone screens explicitly exploratory.

2 RELATED WORK

The AI CUP task uses the four-part formulation described in the PromiseEval shared-task overview: identify an ESG promise, determine whether it has actionable evidence, assess the clarity of the promise–evidence pair, and assign a verification horizon [2]. ML-Promise is the associated multilingual corpus resource; we use its English, French, Japanese, and Korean releases for external replication [16]. This joint verification problem differs from ClimateBERT-NetZero, which detects corporate and national net-zero or reduction targets and supports analysis of their ambition, rather than jointly assessing promise status, evidence, evidence quality, and timing [15].

These dependent outputs form a compact instance of hierarchical text classification, in which labels occupy a supplied structure and coherent predictions respect its ancestor relations [14]. The AI CUP hierarchy comes from the task's official field logic: we neither discover the hierarchy nor propose a new hierarchical architecture. Instead, we compare decision rules over the same field-wise probabilities while preserving or ignoring that given structure.

Yu et al. introduced path-constrained MicroF1 and MacroF1, under which a correct node receives credit only when all its ancestors are also correct [18]. Ji et al. later named this family *C-metrics* [12]. Our path-constrained weighted macro-F1 (C-wF1) adapts Yu et al.'s principle to this task's fields and its official weighted macro-F1; it is therefore not the original C-MacroF1. Section 5.2 defines both it and the set-overlap hierarchical F1 (hF). Both were adopted post hoc and are reported as exploratory.

Plaud et al. observed from point estimates that constrained F1 variants did not globally change method rankings in their experiments [14]. That observation and our post-hoc paired contrast inference answer different questions: ranking a collection of models is not the same estimand as a paired difference between decision rules on shared rows and probabilities. Moreover, our hF changes both consistency treatment and macro-versus-micro aggregation. Of the two post-hoc metrics, only C-wF1 isolates the treatment of children whose predicted ancestors do not support them, while retaining the official field aggregation.

3 TASK AND DATA

Outputs and legal states. The task predicts four fields for each paragraph. We study the AI CUP 2026 VeriPromiseESG development release [1]. Its label inventory and parent–child logic follow PromiseEval [2]. Promise status (PS) is *Yes* or *No*. Verification timeline (VT) is *already*, *within two years*, *between two and five years*, *more than five years*, or *N/A*. Evidence status (ES) is *Yes*, *No*, or *N/A*; evidence quality (EQ) is *Clear*, *Not Clear*, *Misleading*, or *N/A*. These fields obey hard implications: PS=No requires VT=ES=EQ=N/A, whereas PS=Yes requires one of the four substantive timelines and ES in {Yes, No}. In turn, ES=No requires EQ=N/A, whereas ES=Yes requires one of the three substantive quality labels. Thus the Cartesian label space has $2 \times 5 \times 3 \times 4 = 120$ combinations, but only

$$1 + 4(1 + 3) = 17 \tag{1}$$

legal tuples: one for PS=No, and for each of four timelines either one ES=No state or three ES=Yes quality states. We write $\mathcal{S}$ for this set of 17 legal states throughout. Two terms and no others are used for the boundary it draws: a state is *legal* when it belongs to $\mathcal{S}$, and an emitted tuple is *invalid* when it does not.

Table 1: AI CUP 2026 dataset composition and hierarchy audit.

Statistic	Development	Competition test
Paragraphs	2000	2000
Source reports (PDFs)	49	49
shared with other split	49	49
Companies	49	49
Gold labels available	Yes	No
Gold hierarchy-valid tuples	2000 / 2000	n/a
Legal states observed	15 / 17	n/a
Rarest classes		
within_2_years	34	n/a
Misleading	2	n/a

Development-set audit. The release contains 2,000 labeled development paragraphs from 49 source reports and 49 companies. All 2,000 development rows obey the hierarchy, collectively realizing 15 of the 17 legal states (Table 1). The rarest EQ label, *Misleading*, has only two instances; we therefore make no improvement or significance claim for that class.

All 49 development reports also occur in the unlabeled competition test data. Because competition test labels are unavailable, we infer no label-derived test statistic. Each company contributes one report. Consequently, in this corpus a document-disjoint split is also company-disjoint, rather than a separate company-level experiment.

External replication corpora. ML-Promise provides labels in the same four fields [16]. After normalizing its releases to the common 17-state vocabulary, we retain 400 English paragraphs from 9 reports, 400 French paragraphs from 9 reports, and 400 Japanese paragraphs from 19 reports. The Korean release supplies labels, report URLs, and page numbers but no text; we extract the corresponding 500 PDF pages from 32 reports and use the first 384 tokens of each page. The extracted Korean text is not redistributed. Korean has no gold *Misleading* example, and its page-level input is not directly comparable to the paragraph inputs. We therefore interpret only within-corpus paired decision contrasts, never raw score differences between languages.

4 METHODS

4.1 Shared Base Models

For the Chinese study, we use `hfl/chinese-roberta-wwm-ext-large` as the fixed encoder, where `hfl` is the Joint Laboratory of HIT and iFLYTEK Research. BERT-Large has 24 layers and hidden size 1024 [9]; the pinned HFL configuration matches those dimensions. Cui et al. document the checkpoint's Chinese RoBERTa-WWM family: it uses whole-word masking, extended pretraining data, and

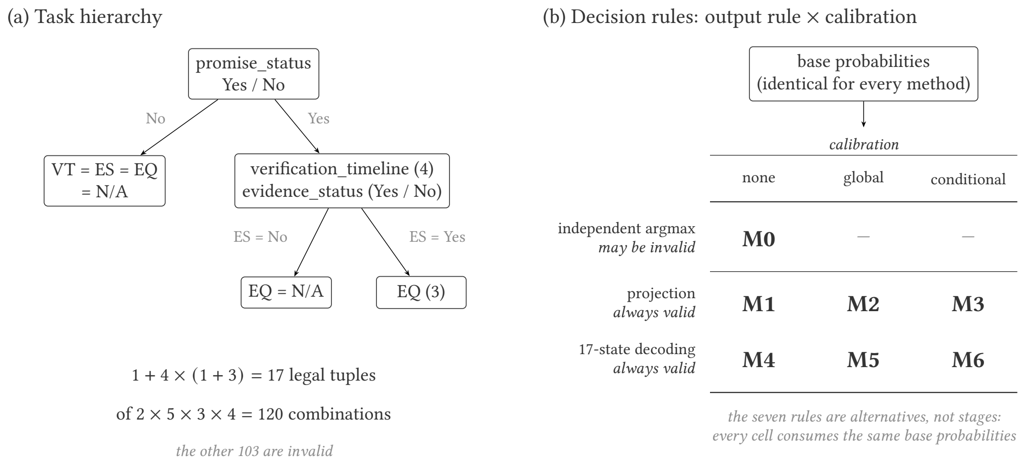

Figure 1: The task hierarchy and the seven decision rules. Panel (b) is the 2×3 design of Section 6: either structured output rule may be combined with any of the three calibration settings, and the cells are alternatives rather than stages. All of them consume identical base probabilities; projection and 17-state decoding guarantee a legal tuple, whereas independent argmax does not, which is why its row has only the uncalibrated cell.

no next-sentence prediction [7]. We mean-pool its token representations and attach four independent linear heads, one per field. The encoder is shared, but neither it nor the heads condition on predicted ancestors. The input is paragraph text only; report URL, company, ticker, and page metadata are excluded.

All fits use the same frozen fine-tuning recipe: maximum length 384, 12 epochs, minibatches of 8 with two-step gradient accumulation (effective size 16), and AdamW with backbone and head learning rates 2×10^{-5} and 10^{-4}. The schedule is cosine with 10% warm-up and layer-wise decay 0.9. The four cross-entropy losses use the official PS/VT/ES/EQ weights 0.20/0.15/0.30/0.35 and label smoothing 0.1. For Train size n, K_t classes present in field t, and class support $n_{t,c}$, the class weight is $\min\{10, n/(K_t\, n_{t,c})\}$ when $n_{t,c} \geq 5$, and 1 otherwise. Hierarchy masking is disabled. We average the last three epoch checkpoints and perform no label-driven early stopping.

For the external fixed arms, English uses RoBERTa-large [13] and French, Japanese, and Korean use XLM-R-large [5]. English robustness runs add large DeBERTa-v3 [10] and ELECTRA [4] checkpoints plus RoBERTa-base. The other languages add RemBERT [3], XLM-R-base, and mBERT [9]. The Chinese anchor adds three exploratory screens of its own, distinct from those English checkpoints: a Chinese DeBERTa-v2-320M [19], a Chinese ELECTRA-large discriminator [6], and the base-sized checkpoint of the primary encoder's own family [7]. Every backbone uses the same maximum length, optimization recipe, three seeds, and five rotations. For each backbone we run the ordinary objective ($\lambda = 0$) and a fixed hierarchy-aware objective ($\lambda = 0.3$):

$$\mathcal{L} = \mathcal{L}_{\text{fields}} - \lambda \log \sum_{s \in \mathcal{S}} \prod_{t} p_{t,s_t}(x). \tag{2}$$

Here $\mathcal{L}_{\text{fields}}$ is the sum of the four weighted cross-entropy losses just described, and $\mathcal{S}$ is the set of 17 legal states of Eq. (1). The second term is the semantic loss of Xu et al. [17]: it penalizes probability mass outside $\mathcal{S}$ without choosing which legal state is correct. Structural-loss and alternate-backbone comparisons are exploratory; the primary decision comparison uses $\lambda = 0$.

4.2 Metric-Aware Decision Calibration

Calibration is the second factor of the comparison: the results table crosses three calibration settings with the two validity-preserving output rules, and each rotation's bias estimates are shared across both rules, which is what lets Table 2 be read as a factorial rather than as separate systems.

For field t and class c, decision calibration adds a class bias in log probability space:

$$z_{t,c}(x) = \log p_{t,c}(x) + b_{t,c}. \tag{3}$$

This changes decision boundaries and is not a claim of probability calibration. Each field's bias vector is fitted by deterministic coordinate ascent with that field's macro-F1 on the held-out Calibration partition as the objective, before either structured output rule is applied. The four objectives therefore decouple. Within every rotation, global and conditional biases are fitted separately; that rotation's global fit is shared by M2/M5 and its conditional fit by M3/M6.

Global biases use every Calibration row. Conditional biases use all rows for PS, only gold PS=Yes rows for VT and ES, and only gold ES=Yes rows for EQ. The three child N/A classes are structurally absent from these subsets. Their biases are therefore unidentifiable and fixed as

$$b_{\text{VT,N/A}} = b_{\text{ES,N/A}} = b_{\text{EQ,N/A}} = 0. \tag{4}$$

This has no effect on projection, which emits N/A only when its parent requires it. It does affect joint scores for (No, N/A, N/A, N/A); hence M6–M5 compares conditional with global estimation including their different estimable parameter sets.

Biases start at zero and are fitted by deterministic coordinate ascent over $[-3, 3]$ in increments of 0.05. Coordinates are visited in the order PS, VT, ES, EQ and, within a field, in the released label order (for EQ: Clear, Not Clear, Misleading, N/A); a move requires a strict improvement, ties favor the smaller absolute bias (then the more negative value), and search stops after a pass without improvement or at 20 passes. Eligibility is determined from the corresponding Train subset: a class absent there is excluded from coordinate ascent and retains bias zero. Among Train-eligible classes, one absent from the matching Calibration subset also retains bias zero and is recorded as an absent-class fallback.

4.3 Validity-Preserving Output Rules

Complete projection. We treat projection as a validity layer over compatible field-wise probabilities, rather than as a new encoder or a claim of improved base-model accuracy. Projection is greedy and top-down. At each field it selects the highest-scoring class allowed by the chosen ancestors. Under PS=No, it sets VT, ES, and EQ to N/A. Under PS=Yes, it excludes N/A from VT and ES and chooses each field's best substantive class. Under ES=No, EQ becomes N/A; under ES=Yes, projection chooses EQ's best substantive class. These rules completely enforce both directions of every parent implication. This bidirectional projection always returns one of the 17 legal states; a one-way N/A overwrite would not.

17-state decoding. The alternative scores every $s \in \mathcal{S}$ and returns

$$\hat{y} = \arg\max_{s \in \mathcal{S}} \sum_t \{\log p_{t,s_t}(x) + b_{t,s_t}\}. \quad (5)$$

The four fields enter this sum unweighted. A per-field scale α_t would be the obvious generalization, and we deliberately do not fit one: the comparison against projection is meant to isolate the choice of output rule, and a decoder carrying four tuned parameters that projection does not have would no longer answer that question.

Figure 1 summarizes seven *alternative* rules over identical cross-fitted base probabilities, not sequential stages: M0 is uncalibrated independent argmax; M1/M2/M3 use projection with respectively no, global, or conditional bias; and M4/M5/M6 use 17-state decoding with the same three bias choices. Only M0 can emit an invalid tuple.

5 EXPERIMENTAL DESIGN

We first define the shared evaluation protocols and inference. Table 2 then summarizes the controlled comparison interpreted in Section 6.

5.1 Cross-Fitted Evaluation Targets

We use five-way rotating three-way cross-fitting. For rotation k, Test is fold k, Calibration is fold $(k+1) \bmod 5$, and Train comprises the other three folds; the three partitions are mutually exclusive. After five rotations, every development paragraph has served as Test exactly once. We concatenate the 2,000 out-of-fold predictions and compute each metric once, rather than average fold-level F1 values whose present gold-label sets may differ. The three seeds 42, 123, and 456 each determine both a fold assignment and training randomness, so seed variation represents the complete pipeline.

The primary, document-disjoint protocol balances report row counts with a randomized greedy procedure. After a seeded shuffle, it processes reports from largest to smallest and randomly places each in one of the two currently lightest folds. It estimates unseen-report generalization and, given the corpus audit, is also company-disjoint.

The same-document protocol instead jointly stratifies by gold PS, ES, and VT. It shuffles rows within each joint-label bucket and deals them round-robin to the five folds. Every report represented in Calibration or Test must also occur in Train through a different paragraph. This estimates seen-report, unseen-paragraph generalization. The two protocols target distinct estimands; neither is a biased and the other an unbiased version of one quantity.

Within each protocol and seed, M0–M6 consume the same Test rows and base probabilities.

External runs use document-disjoint report grouping, three seeds, and five rotations. Each language has eight paired arms – four backbones under both $\lambda = 0$ and 0.3 – giving 32 external arms, none of them chosen on an observed score. The same-document protocol is descriptive and available only for the primary large backbone. The complete executed study contains 765 training artifacts and 1,050 row-level prediction files across five corpora.

5.2 Metrics and Inference

The field-wise weighted macro-F1 is a standard choice for multi-field classification, but it contains no term that assesses the joint validity of a four-field tuple. For Chinese, the primary metric is the official weighted macro-F1,

$$\text{wF1} = 0.20F_{\text{PS}} + 0.15F_{\text{VT}} + 0.30F_{\text{ES}} + 0.35F_{\text{EQ}}, \quad (6)$$

where each F_t is field-level macro-F1 over classes present in the gold labels being scored. Because this denominator depends on gold support, per-class leverage differs by corpus: a unit gain on one EQ class contributes $0.35/3 = 0.1167$ to the Korean weighted total but $0.35/4 = 0.0875$ elsewhere. Pre-specified whole-tuple accuracy requires all four fields to match. Path-constrained weighted macro-F1 (C-wF1) and hierarchical F1 (hF) were adopted post hoc. C-wF1 retains the official field weights but counts a substantive child prediction as wrong when its predicted ancestors do not support it. Following the hierarchical precision/recall/F1 set-overlap definition [14], our hF micro-averages over emitted non-N/A field–value nodes: hP is total gold–prediction overlap divided by predicted nodes, hR divides the same overlap by gold nodes, and hF is their harmonic mean. For an inconsistent emitted tuple, we do not add missing ancestors before scoring; legal M1–M6 are unaffected because the conventions coincide for them. hF changes both consistency treatment and macro-versus-micro averaging, so the two supplementary metrics answer different questions.

ML-Promise has no official weighted metric. We therefore label its scores "AI CUP weights applied to ML-Promise." All external comparisons remain paired within corpus, backbone, and λ.

For paired method contrasts, we perform a paired PDF-cluster bootstrap over 49 source-report clusters. Each of 10,000 replicates samples 49 reports with replacement using seed 20260814, applies the same sampled rows to both methods and all three seeds, computes the within-seed differences, and averages those differences. The five pre-specified contrasts are M1–M0, M3–M2, M4–M1,

Table 2: Document-disjoint cross-fitted results on the Chinese development set.

ID	Calibration	Decoding	wF1 (official)	PS	VT	ES	EQ	Tuple acc.	Invalid %
M0	None	Independent	0.572±0.003	0.796	0.464	0.650	0.424	0.359±0.007	12.55
M1	None	Projection	0.571±0.003	0.796	0.467	0.651	0.418	0.394±0.004	0.00
M2	Global	Projection	0.574±0.004	0.796	0.493	0.650	0.418	0.428±0.004	0.00
M3	Conditional	Projection	0.575±0.003	0.796	0.501	0.651	0.416	0.432±0.004	0.00
M4	None	17-state	0.571±0.005	0.799	0.468	0.648	0.420	0.388±0.003	0.00
M5	Global	17-state	0.576±0.001	0.796	0.493	0.649	0.423	0.430±0.001	0.00
M6	Conditional	17-state	0.567±0.008	0.784	0.494	0.634	0.418	0.427±0.005	0.00

Table 3: Chinese paired contrasts on the two pre-specified metrics. Official wF1 and tuple accuracy are separate Holm families of five. Intervals are uncorrected 95% report-cluster bootstrap intervals. Post-hoc C-wF1 and hF are discussed only in the text.

Metric	Contrast	Δ	95% CI	p_{Holm}
wF1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.416
	M6-M5	-0.008	[-0.016, -0.001]	0.171
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.004	[-0.001, 0.008]	0.370
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.015, 0.004]	0.505
	M6-M5	-0.004	[-0.011, 0.004]	0.505

M6–M3, and M6–M5. M1–M0 changes only the output rule, with calibration held out of both arms; it is therefore the cleanest test of projection rather than the highest-scoring configuration. Our only confirmatory family applies Holm correction to these five contrasts on Chinese official wF1 [11]. Pre-specified Chinese tuple exact-match is a separate secondary family of five. Bracketed 95% intervals are uncorrected bootstrap percentile intervals; decisions use $p_{\text{Holm}} < .05$, not whether an interval excludes zero.

External replication was pre-registered around within-arm contrast directions; we additionally apply a report-cluster bootstrap only to the fixed English M1–M0 tuple contrast. We do not pool languages into a post-hoc significance test. C-wF1 and hF were added after the Chinese primary result, while structural-loss and alternate-backbone screens test robustness; all are exploratory and do not support confirmatory claims regardless of their nominal p-values.

6 RESULTS

Guaranteed validity and Chinese confirmatory evidence. Table 2 compares decision rules on identical base probabilities and Test rows. Independent argmax (M0) produces invalid tuples for 12.55% of document-disjoint predictions; projection and 17-state decoding (M1–M6) reduce this rate to zero by construction. This zero rate is a deterministic property of the validity layer, not evidence that the base model became more accurate. Pooled over both protocols and all three seeds, M0 emits 1,527 invalid tuples in 12,000 rows,

Table 4: Projection (M1) versus independent argmax (M0) on five corpora, each using its preselected primary backbone at $\lambda = 0$ and document-disjoint evaluation; values are means over three seeds. M1 guarantees 0% invalid output. The *N/A* and substantive net columns summarize changes in correct child predictions after projection. Official-score deltas should not be compared directly across corpora because macro-F1 averages over gold-present classes, and Korean contains no Misleading example.

Corpus	Rows	M0 invalid %	*N/A* net	Subst. net	Δ wF1	Δ tuple acc.
Chinese (AI CUP)	2,000	12.55	+395	-264	-0.0011	**+0.0350**
English	400	23.17	+148	-86	+0.0032	**+0.0725**
French	400	21.08	+150	-107	+0.0036	**+0.0625**
Japanese	400	31.25	+183	-245	-0.0131	**+0.0708**
Korean	500	22.73	+103	-148	-0.0192	**+0.0293**

Table 5: Anatomy of the 1,527 invalid tuples independent argmax (M0) emits over 12,000 scored rows – both protocols, all three seeds. In (a) a row is attributed to the first implication it breaks, so the four counts partition the 1,527 exactly. Panel (b) derives the Introduction's motivating figure: restricted to those same invalid rows, each field is scored on its own, so the row still collects its weight times its per-field accuracy, and the four terms sum to 0.443. That is weighted field accuracy on invalid rows, not weighted macro-F1, which has no per-row decomposition.

(a) Which implication the invalid tuple breaks		
Violated implication	Rows	Share
PS=No, some child not *N/A*	699	45.8%
ES=No, EQ substantive	755	49.4%
PS=Yes, VT or ES *N/A*	47	3.1%
ES=Yes, EQ=*N/A*	26	1.7%
All invalid tuples	1,527	100.0%
(b) Weighted field credit those rows still collect		
Field	Weight × accuracy	Credit
PS	0.20 × 0.672	0.134
VT	0.15 × 0.320	0.048
ES	0.30 × 0.360	0.108
EQ	0.35 × 0.436	0.153
Total	1.00	0.443

of which 1,454 (95.2%) break one of two implications: a substantive child under PS=No, or an evidence quality under ES=No. Table 5(a) gives the full breakdown, and Table 5(b) derives the 44.3% weighted field credit those rows retain. The failure is concentrated in predictions that deny a parent and then answer its children anyway, rather than spread evenly over the label space. The ± values are sample standard deviations across the three seeds, on both the official metric and tuple accuracy. Because each seed determines both fold assignment and training randomness, they describe spread over the whole split-and-training pipeline, not a confidence interval.

Table 3 reports both pre-specified families in full. All five official wF1 contrasts are undetected after Holm correction. For the central M1–M0 contrast, $\Delta = -0.001$ with 95% interval $[-0.006, 0.003]$ and $p_{\text{Holm}} = 1.000$. This is the absence of a detected difference; it does not establish a zero effect.

The null is not confined to that contrast. The seven decision rules span 0.008 in official wF1, and that spread is itself a single contrast: M5 is the highest-scoring rule and M6 the lowest, so the range equals |M6 − M5|, which has an interval excluding zero ($[-0.016, -0.001]$) and still does not survive correction (p_{Holm} = .171). The largest gap between any two of the rules compared here is therefore below what this evaluation can resolve, which bounds what any decision-stage comparison on this benchmark can demonstrate.

Whole-tuple effects. Exact whole-tuple accuracy tells a different story. M1–M0 is +0.035 $[0.028, 0.043]$, p_{Holm} = .001. Conversely, 17-state decoding is not uniformly better: M4–M1 is -0.006 $[-0.010, -0.002]$, p_{Holm} = .028. Searching every legal state can therefore reduce whole-row correctness relative to greedy projection. The validity guarantee is shared, but the choice among legal tuples still matters.

The two calibration contrasts are also undetected: conditional bias, fitted only on the subset each parent admits, is not distinguishable from global bias under either output rule (M3–M2 and M6–M5, on both pre-specified metrics). The hierarchy therefore constrains the output rule usefully without improving how the biases are estimated. The calibrated projection arms reach 0.428 and 0.432 on tuple accuracy against M1's 0.394, a step no pre-specified contrast covers.

The post-hoc C-wF1 and hF M1–M0 point estimates are also positive (+0.004 and +0.003) but do not alter the inferential hierarchy; under same-document evaluation C-wF1 stays positive at +0.002, on an interval $[-0.001, 0.005]$ that crosses zero.

Evaluation protocol. Section 5 treats the two protocols as targeting distinct estimands rather than as a biased and an unbiased version of one quantity; Table 6 is what that separation costs. Every rule scores higher under same-document evaluation, by 0.010 to 0.020, and every interval excludes zero. The gap is therefore a property of the split rather than of any rule, and it is larger than the 0.008 that separates the rules themselves: on this benchmark, which paragraphs are held out matters more than which decision rule is applied to them.

Multilingual replication. Table 4 fixes one arm per language before inspecting scores, each under document-disjoint evaluation at $\lambda = 0$. Whole-tuple accuracy rises in all five corpora, whereas weighted macro-F1 rises only for English and French; Section 7.1 reads its two ledger columns.

Table 7 reports all four external corpora at both loss settings, arm by arm. M1–M0 tuple accuracy is positive in 32/32 arms. The weighted macro-F1 contrast is positive in only 14/32: English 7/8, French 5/8, Japanese 0/8, and Korean 2/8. These are within-arm directions, not a raw score ranking across languages. Projection thus combines a deterministic validity guarantee with a repeated complete-tuple improvement, while its field-score effect is not directionally stable across arms.

Table 6: Chinese anchor scored under both evaluation protocols. Same-document holds out paragraphs from reports seen in training; document-disjoint holds out whole reports. Δ separates two estimands and is not a bias estimate. Columns are rounded independently from the unrounded scores, so Δ can differ by 0.001 from the difference of the two columns beside it.

Method	Same-document	Document-disjoint	Δ	95% CI
M0	0.585	0.572	0.012	[0.004, 0.022]
M1	0.581	0.571	0.010	[0.001, 0.019]
M2	0.587	0.574	0.012	[0.001, 0.024]
M3	0.588	0.575	0.012	[0.002, 0.023]
M4	0.586	0.571	0.014	[0.005, 0.024]
M5	0.590	0.576	0.015	[0.004, 0.027]
M6	0.587	0.567	0.020	[0.011, 0.030]

For the pre-registered fixed English arm, the one-sided tuple effect is +0.0725, with a report-cluster bootstrap interval of $[0.0452, 0.1019]$ and p = .0002. That p is resampled from nine report clusters, which is the resolution the English corpus has: it should be read as a direction that survived its own resampling, not as a precise tail probability. ML-Promise weighted scores use AI CUP weights applied to ML-Promise.

Exploratory robustness. On the Chinese anchor, structural loss lowers M0's invalid-tuple rate from 12.55% to 5.18%; its pre-registered M1 wF1 effect is +0.00250 $[-0.00384, 0.00860]$ (p = .427). It also lowers that rate in all 16 external language-by-backbone pairs, with mixed wF1 and tuple effects. The Chinese DeBERTa and ELECTRA screens likewise lack conclusive wF1 gains; the same-family base checkpoint's lower invalid-tuple rate contradicts a simple smaller-model mechanism.

Table 8 reports those seven Chinese arms as paired within-arm contrasts. Its invalid-rate column shows what the penalty does not do: the $\lambda = 0.3$ arms still emit invalid tuples, because the training-time penalty pulls probability mass toward $\mathcal{S}$ without confining the output to $\mathcal{S}$, which is what a decision rule does. Two sign patterns then hold across all seven arms even though the official column does not: M1–M0 raises tuple accuracy in 7/7, and M4–M1 lowers it in 7/7. The single-arm M4–M1 result above is the sign the whole screen shows, not one arm's outcome.

7 DISCUSSION

The practical contribution of projection is its guarantee rather than a universal field-score gain: as a retraining-free layer over compatible field-wise probabilities, it enforces the supplied hierarchy in every arm and improves exact whole-tuple accuracy throughout the executed external matrix, without any claim about the underlying encoder.

Table 7: Every executed external arm behind the 32/32 and 14/32 summaries. Each cell is the M1−M0 within-arm contrast under document-disjoint evaluation, averaged over three seeds; the arms were fixed before any score was inspected. Bold marks a positive tuple-accuracy contrast. Four decimals, because three would print several arms as ±0.000 and hide the sign the counts are made of. Backbones are corpus-prefixed; the identically named checkpoints in Table 8 are Chinese. ML-Promise has no official weighted metric, so its wF1 applies AI CUP weights.

		$\lambda = 0$			$\lambda = 0.3$		
Corpus	Backbone	M0 inv. %	ΔwF1	Δtuple	M0 inv. %	ΔwF1	Δtuple
English	RoBERTa-large	23.17	+0.0032	**+0.0725**	13.25	+0.0033	**+0.0350**
	DeBERTa-v3-large	17.25	-0.0044	**+0.0350**	8.92	+0.0029	**+0.0175**
	ELECTRA-large	17.08	+0.0030	**+0.0400**	7.50	+0.0044	**+0.0233**
	RoBERTa-base	20.42	+0.0001	**+0.0500**	9.33	+0.0012	**+0.0217**
French	XLM-R-large	21.08	+0.0036	**+0.0625**	9.58	+0.0025	**+0.0300**
	RemBERT	21.67	-0.0117	**+0.0442**	8.92	-0.0021	**+0.0183**
	XLM-R-base	27.25	+0.0004	**+0.0650**	12.83	-0.0054	**+0.0283**
	mBERT	21.50	+0.0025	**+0.0483**	9.50	+0.0024	**+0.0150**
Japanese	XLM-R-large	31.25	-0.0131	**+0.0708**	8.58	-0.0077	**+0.0108**
	RemBERT	40.67	-0.0171	**+0.0625**	13.58	-0.0042	**+0.0142**
	XLM-R-base	37.08	-0.0043	**+0.0733**	10.67	-0.0100	**+0.0125**
	mBERT	34.67	-0.0196	**+0.0733**	8.75	-0.0060	**+0.0175**
Korean	XLM-R-large	22.73	-0.0192	**+0.0293**	8.87	-0.0032	**+0.0147**
	RemBERT	32.07	-0.0164	**+0.0513**	10.27	+0.0015	**+0.0207**
	XLM-R-base	28.93	-0.0317	**+0.0367**	10.07	-0.0047	**+0.0127**
	mBERT	21.80	-0.0245	**+0.0220**	8.07	+0.0008	**+0.0140**

Table 8: Chinese backbone and structural-loss screens, as paired within-arm contrasts. All checkpoints are Chinese; the base-sized arm was pre-registered without a training-objective condition, so it appears at $\lambda = 0$ only. Bold marks a contrast whose uncorrected 95% paired report-cluster bootstrap interval excludes zero; four more interval columns would not fit this width, so the intervals are released with the data rather than printed. These contrasts are exploratory and form no Holm family.

			Enforce legality (M1−M0)		Joint decoding (M4−M1)	
Backbone	λ	M0 inv. %	wF1	Tuple acc.	wF1	Tuple acc.
Chinese RoBERTa-large	0	12.55	-0.001	**+0.035**	+0.000	**-0.006**
Chinese DeBERTa-v2-320M	0	19.75	**-0.008**	**+0.050**	**+0.013**	-0.003
Chinese ELECTRA-large	0	10.20	**-0.005**	**+0.024**	**+0.007**	-0.000
Chinese RoBERTa-base	0	11.77	**-0.007**	**+0.030**	**+0.008**	-0.003
Chinese RoBERTa-large	0.3	5.18	-0.001	**+0.014**	-0.001	-0.001
Chinese DeBERTa-v2-320M	0.3	6.53	+0.001	**+0.015**	-0.002	**-0.003**
Chinese ELECTRA-large	0.3	3.68	-0.000	**+0.009**	-0.000	-0.002

7.1 Why the two metrics disagree

This guarantee is not cost-free in every scoring view, and Table 4 makes the trade-off explicit by counting the correct field predictions projection creates against those it destroys. The *N/A* net is positive and the substantive-class net negative in all five fixed arms. If an independently predicted parent is wrong, projection may replace a correct substantive child with the structurally required *N/A*; when the parent is correct, the same operation can repair an unsupported child. Field-wise macro-F1 values these changes class by class, whereas tuple accuracy credits only four-field agreement. The field-score effect therefore varies across backbones and corpora. Post-hoc C-wF1 and hF point in the tuple direction, but were chosen after the primary analysis and are not needed to establish the pre-specified secondary result.

Validity alone also does not identify the better structured rule. In Chinese, M4 scores all 17 legal states jointly yet loses tuple accuracy to greedy M1. Joint search can improve one field while changing another field that projection had already made correct. This adverse result cautions against treating a larger legal-state search as automatically superior; it does not imply that joint decoding must fail on other models or tasks.

7.2 Reproducibility

Code, pre-registration documents, row-level predictions for all five corpora, and the pipeline that regenerates every table here are at https://github.com/EXAMPLE/REPO-URL-TBD.

Checkpoints. Every backbone is a pinned Hugging Face checkpoint, its commit revision recorded in the released configuration. Chinese: `hfl/chinese-roberta-wwm-ext-large` (anchor), `hfl/chinese-roberta-wwm-ext`, `IDEA-CCNL/Erlangshen-DeBERTa-v2-320M-Chinese`, `hfl/chinese-electra-180g-large-discriminator`. English: `roberta-large`, `roberta-base`, `microsoft/deberta-v3-large`, `google/electra-large-discriminator`. French, Japanese and Korean: `FacebookAI/xlm-roberta-large`, `FacebookAI/xlm-roberta-base`, `google/rembert`, `google-bert/bert-base-multilingual-cased`.

Procedures given only in outline above. The document-disjoint split sorts reports by identifier, shuffles them under the seed, re-sorts by descending row count with a stable sort so the shuffle survives ties, then assigns each report uniformly at random to one of the two currently lightest folds, ties broken by fold index. Korean pages come from Poppler's `pdftotext-fP-lP-layout`, one page per labelled (URL, page) pair, falling back to `pdftoppm-r250` and Tesseract (`-lkor+eng`) for image-only pages; the release already carries one label tuple per unique (URL, page), so no further deduplication applies. Python 3.10.20, PyTorch 2.2.2+cu121 and Transformers 4.40.2 on Linux; the 32 external arms consumed 25.4 GPU-hours in total on a single RTX 3090.

7.3 Limitations

The corpora differ in language, reports, annotation distributions, and primary backbones, so the replication matrix is not a factorial estimate of a pure language effect. English and French each have only nine report clusters, and even Chinese inference resamples just 49. The 32/32 and 14/32 summaries are therefore descriptive directions across executed arms, not a pooled significance test. AI CUP weights applied to ML-Promise are not an official ML-Promise metric.

Every contrast here is measured against M0, the unconstrained argmax over our own probabilities. That is the right comparator for the question we ask, since it holds rows and probabilities fixed, but the paper therefore reports no comparison against a published system, an official baseline, or a leaderboard, and nothing here is a claim about where this system ranks among participants.

Korean has a further input mismatch: its 500 examples are first-384-token PDF pages rather than released paragraphs, and it has no gold *Misleading* example. We do not redistribute the extracted Korean text or source PDFs. The reported Korean predictions are released and can therefore be rescored, but retraining requires rebuilding the page inputs from their source documents with the recipe in Section 7.2.

Results for structural loss and other backbones remain exploratory: they show that reducing the invalid-tuple rate need not yield a conclusive gain on the official score, but the fixed loss weights and limited checkpoint set cannot characterize all training objectives or architectures. The rarest Chinese quality label has two examples and Korean has none, so no class-level claim is licensed for it. Finally, each seed jointly changes folds and optimization randomness, so those two sources of variation cannot be separated.

8 CONCLUSION

Hierarchical projection is not a guaranteed route to higher field-wise weighted macro-F1. It is instead a retraining-free validity layer that eliminates invalid tuples by construction and improves exact whole-tuple accuracy on the Chinese anchor and in all 32 evaluated external arms. Its field-score effect varies across models and corpora, but its logical guarantee does not. For audit-oriented tasks with dependent outputs, field-wise performance, whole-tuple correctness, and structural validity should be reported together.

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